Yusuf Kaan Coşar Room 2 150719569

On my honor, I have neither given nor received any unauthorized and/or inappropriate asistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

1-

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 1 & \frac{5}{3\alpha} & \frac{-3\gamma}{\alpha} & \frac{1}{3}\alpha \\ 1 & \frac{\beta}{2} & \frac{3}{\alpha} & \frac{\alpha^2}{2} \\ 1 & 14 & -6 & 2 \end{bmatrix} \xrightarrow[r_3-r_1\to r_3]{r_2-r_1\to r_2} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & \frac{2}{3\alpha} & \frac{-3\gamma+\gamma\alpha}{\alpha} & -\frac{2}{3}\alpha \\ 0 & \frac{\beta}{2}-\alpha & \frac{3}{2}+\gamma & \frac{\alpha^2}{2}-\alpha \\ 0 & 14-\alpha & -6+\gamma & 2-\alpha \end{bmatrix}$$

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2 - inci dimensional space $x = 5y$

a-) En az 3 boyutludur. Çünkü hacim 3 boyutludur.

b)

c) $x + 4y - z = 6$

$x - 5y = 0$

$$\left[\begin{array}{ccc|c} 1 & 4 & -1 & 6 \\ 1 & -5 & 0 & 0 \end{array}\right] \xrightarrow{r_2 - r_1} \left[\begin{array}{ccc|c} 1 & 4 & -1 & 6 \\ 0 & -9 & 0 & 0 \end{array}\right] \xrightarrow{-\frac{r_2}{9} \to r_2} \left[\begin{array}{ccc|c} 1 & 4 & -1 & 6 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{-4r_2 + r_1} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 6 \\ 0 & 1 & 0 & 0 \end{array}\right]$$

2D inside 2D

$y = 0$

$x - z = 6$

Evet vektör uzayıdır

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3- Symetric matrix $H = I - 2uu^T$

a-) $H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

$H^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T$

$H^2 \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

b) Bulunduğu uzayı gerer
Kümenin elemanları lineer bağımsız.
Birbirlerine dikler

Bu yüzden ortogonaller

c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix}$

$(1-\lambda)^2 = 0$

$\boxed{\lambda = 1}$

$\lambda = 1$ $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0$

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

d) Köşegenleştirilebilir. Çünkü n tane bağımsız vektör vardır.

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4-)

a) Bence bize verilen sayıların denklem doğru

b) $\begin{vmatrix} 2-\lambda & -2 \\ -1 & 1-\lambda \end{vmatrix} = 0$

$(2-\lambda)(1-\lambda) - 2$

$2 - 2\lambda - \lambda + \lambda^2 - 2$

$\lambda^2 - 3\lambda = 0$

$\lambda = 0$ $\lambda = 3$

$\lambda = 0$

$\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda = 3$

$\begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0 \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$c_1 \cdot \lambda_1^n \cdot X_1 + c_2 \cdot \lambda_2^n \cdot X_2$

$c_1 \cdot 0^n \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 3^n \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$c_1 = \frac{1}{3}$ $c_2 = \frac{1}{3}$

$\frac{1}{3} 0^1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{3} \cdot 3^1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

c) $3^{n \to 0}$

$\begin{bmatrix} x_\infty \\ y_\infty \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$66.6\overline{6}$

$33.3\overline{3}$

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